

Exercise 1: Configuring a Basic Spring Application

pom.xml

```
<dependencies>

    <!-- Spring Context -->
    <dependency>
        <groupId>org.springframework</groupId>
        <artifactId>spring-context</artifactId>
        <version>5.3.39</version>
    </dependency>

</dependencies>
```

BookRepository.java

(Package: com.library.repository)

```
package com.library.repository;

public class BookRepository {

    public void displayBook() {
        System.out.println("BookRepository: Fetching book details...");
    }
}
```

BookService.java

(Package: com.library.service)

```
package com.library.service;

import com.library.repository.BookRepository;

public class BookService {

    private BookRepository bookRepository;

    // Setter Injection
    public void setBookRepository(BookRepository bookRepository) {
```

```

        this.bookRepository = bookRepository;
    }

    public void showBook() {
        System.out.println("BookService: Calling repository...");
        bookRepository.displayBook();
    }
}

```

applicationContext.xml

(Place in src/main/resources)

```

<?xml version="1.0" encoding="UTF-8"?>

<beans xmlns="http://www.springframework.org/schema/beans"
       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
       xsi:schemaLocation="
           http://www.springframework.org/schema/beans
           https://www.springframework.org/schema/beans/spring-beans.xsd">

    <!-- Repository Bean -->
    <bean id="bookRepository"
          class="com.library.repository.BookRepository"/>

    <!-- Service Bean -->
    <bean id="bookService"
          class="com.library.service.BookService">

        <property name="bookRepository"
                  ref="bookRepository"/>

    </bean>

</beans>

```

MainApp.java

(Package: com.library)

```

package com.library;

import com.library.service.BookService;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;

public class MainApp {

    public static void main(String[] args) {

        ApplicationContext context =
            new ClassPathXmlApplicationContext("applicationContext.xml");

        BookService service = context.getBean("bookService", BookService.class);

        service.showBook();
    }
}

```



```

Run Main x
"C:\Program Files\Java\jdk-17\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.2.1\lib\idea_rt.jar=64551" -Dfile.encoding=UTF-8 -classpath D:\LibraryManagement-Assignment\target\classes
BookService: Calling repository...
BookRepository: Fetching book details...
Process finished with exit code 0

```

Exercise 2: Implementing Dependency Injection

pom.xml

```

<dependencies>

    <!-- Spring Context -->
    <dependency>
        <groupId>org.springframework</groupId>
        <artifactId>spring-context</artifactId>
        <version>5.3.39</version>
    </dependency>

```

</dependencies>

applicationContext.xml

<?xml version="1.0" encoding="UTF-8"?>

<beans xmlns="http://www.springframework.org/schema/beans"

xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"

xsi:schemaLocation="

http://www.springframework.org/schema/beans

https://www.springframework.org/schema/beans/spring-beans.xsd">

<bean id="bookRepository"

class="com.library.repository.BookRepository"/>

<bean id="bookService"

class="com.library.service.BookService">

<property name="bookRepository"

ref="bookRepository"/>

</bean>

</beans>

BookRepository.java

package com.library.repository;

```
public class BookRepository {

    public void displayBook() {
        System.out.println("BookRepository: Fetching book details...");
    }

}
```

BookService.java

```
package com.library.service;

import com.library.repository.BookRepository;

public class BookService {

    private BookRepository bookRepository;

    public void setBookRepository(BookRepository bookRepository) {
        this.bookRepository = bookRepository;
    }

    public void showBook() {
        System.out.println("BookService is working...");
        bookRepository.displayBook();
    }

}
```

LibraryManagementApplication.java

```

package com.library;

import com.library.service.BookService;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;

public class LibraryManagementApplication {

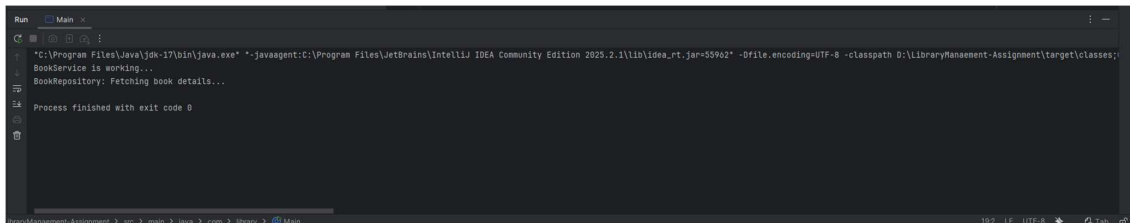
    public static void main(String[] args) {

        ApplicationContext context =
            new ClassPathXmlApplicationContext("applicationContext.xml");

        BookService service =
            context.getBean("bookService", BookService.class);

        service.showBook();
    }
}

```



Exercise 4: Creating and Configuring a Maven Project

Pom.xml

```

<?xml version="1.0" encoding="UTF-8"?>

<project xmlns="http://maven.apache.org/POM/4.0.0"

    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"

    xsi:schemaLocation="http://maven.apache.org/POM/4.0.0

        http://maven.apache.org/xsd/maven-4.0.0.xsd">

```

<modelVersion>4.0.0</modelVersion>

<groupId>com.library</groupId>

<artifactId>LibraryManagement</artifactId>

<version>1.0-SNAPSHOT</version>

<properties>

 <maven.compiler.source>1.8</maven.compiler.source>

 <maven.compiler.target>1.8</maven.compiler.target>

 <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>

</properties>

<dependencies>

 <!-- Spring Context -->

 <dependency>

 <groupId>org.springframework</groupId>

 <artifactId>spring-context</artifactId>

 <version>5.3.39</version>

 </dependency>

 <!-- Spring AOP -->

 <dependency>

 <groupId>org.springframework</groupId>

 <artifactId>spring-aop</artifactId>

 <version>5.3.39</version>

 </dependency>

```
<!-- Spring Web MVC -->

<dependency>

  <groupId>org.springframework</groupId>

  <artifactId>spring-webmvc</artifactId>

  <version>5.3.39</version>

</dependency>

</dependencies>

<build>

  <plugins>

    <plugin>

      <groupId>org.apache.maven.plugins</groupId>

      <artifactId>maven-compiler-plugin</artifactId>

      <version>3.11.0</version>

      <configuration>

        <source>1.8</source>

        <target>1.8</target>

      </configuration>

    </plugin>

  </plugins>

</build>

</project>
```

External Libraries

- > < 17 > C:\Program Files\Java\jdk-17
- > Maven: org.springframework:spring-aop:5.3.39
- > Maven: org.springframework:spring-beans:5.3.39
- > Maven: org.springframework:spring-context:5.3.39
- > Maven: org.springframework:spring-core:5.3.39
- > Maven: org.springframework:spring-expression:5.3.39
- > Maven: org.springframework:spring-jcl:5.3.39
- > Maven: org.springframework:spring-web:5.3.39
- > Maven: org.springframework:spring-webmvc:5.3.39